

### 5.3.6.1 Current consumption in Run mode

**Table 23. Typical and maximum current consumption in Run mode**

Evaluated by characterization - not tested in production, unless otherwise stated.

Clocked by HSI at 144 MHz or HSIDIV3 at 48MHz if not otherwise specified.

| Symbol                              | Parameter                                                 | Conditions                          | f <sub>HCLK</sub><br>(MHz) | Typ  | Max                      |                          |                           |                           |                           | unit |
|-------------------------------------|-----------------------------------------------------------|-------------------------------------|----------------------------|------|--------------------------|--------------------------|---------------------------|---------------------------|---------------------------|------|
|                                     |                                                           |                                     |                            |      | T <sub>J</sub> =<br>30°C | T <sub>J</sub> =<br>90°C | T <sub>J</sub> =<br>110°C | T <sub>J</sub> =<br>130°C | T <sub>J</sub> =<br>140°C |      |
| I <sub>DD(Run)</sub> <sup>(1)</sup> | Supply current in Run mode all peripheral clocks disabled | Code in Flash, ICACHE 2-Ways        | 144                        | 11.5 | 12.00                    | 13.50                    | 14.50                     | 17.00                     | 18.50                     | mA   |
|                                     |                                                           |                                     | 48                         | 4.1  | 4.35                     | 5.75                     | 7.15                      | 9.45                      | 11.00                     |      |
|                                     |                                                           | Code in Flash, ICACHE OFF           | 144                        | 9.4  | 10.00                    | 11.50                    | 12.50                     | 15.00                     | 16.50                     |      |
|                                     |                                                           |                                     | 48                         | 4.0  | 4.35                     | 5.70                     | 7.05                      | 9.35                      | 11.00                     |      |
|                                     |                                                           | Code in SRAM2, ICACHE OFF, FLASH ON | 144                        | 9.3  | 9.80                     | 11.00                    | 12.50                     | 15.00                     | 16.50                     |      |
|                                     |                                                           |                                     | 48                         | 3.4  | 3.60                     | 5.05                     | 6.40                      | 8.70                      | 10.50                     |      |
|                                     | Supply current in Run mode all peripheral clocks enabled  | Code in Flash, ICACHE 2-Ways        | 144 <sup>(2)</sup>         | 23.5 | 24.50                    | 25.50                    | 27.00                     | 29.50                     | 31.50                     |      |
|                                     |                                                           | Code in Flash, ICACHE OFF           |                            | 21.0 | 22.50                    | 23.50                    | 25.00                     | 27.50                     | 29.00                     |      |

1. Measures done with prefetch enabled.

2. Clocked by PSI at 144 MHz with HSE at 16 MHz in bypass mode.

**Table 24. Typical current consumption in Run mode with CoreMark running from flash memory and SRAM**

Evaluated by characterization - not tested in production, unless otherwise stated.

| Symbol                              | Parameter                  | Conditions                               | SYSCLK source      | f <sub>HCLK</sub><br>(MHz) | Typ   | Unit | Typ   | Unit    |
|-------------------------------------|----------------------------|------------------------------------------|--------------------|----------------------------|-------|------|-------|---------|
|                                     |                            | Peripheral                               |                    |                            |       |      |       |         |
| I <sub>DD(Run)</sub> <sup>(1)</sup> | Supply current in Run mode | Code in Flash, ICACHE 2-WAY, prefetch ON | PSI <sup>(2)</sup> | 144                        | 12.00 | mA   | 82.00 | μA/ MHz |
|                                     |                            | Code in Flash, ICACHE 1-WAY, prefetch ON |                    |                            | 9.85  |      | 68.50 |         |
|                                     |                            | Code in Flash, ICACHE OFF, prefetch ON   |                    |                            | 9.45  |      | 65.50 |         |
|                                     |                            | Code in Flash, ICACHE OFF, prefetch OFF  |                    |                            | 8.40  |      | 58.50 |         |
|                                     |                            | Code in SRAM2, ICACHE 2-WAY              |                    |                            | 11.50 |      | 78.00 |         |
|                                     |                            | Code in SRAM2, ICACHE 1-WAY              |                    |                            | 9.35  |      | 65.00 |         |
|                                     |                            | Code in SRAM2, ICACHE OFF                |                    |                            | 9.15  |      | 63.50 |         |

1. Measures done with prefetch enabled.

2. Clocked by PSI 144 MHz with HSE at 16 MHz on bypass mode.

### 5.3.6.2 Current consumption in Sleep mode

**Table 25. Typical and maximum current consumption in Sleep mode**

Evaluated by characterization - Not tested in production.

Clocked by HSI at 144MHz or HSIDIV3 at 48MHz.

| Symbol                 | Parameter                    | Conditions                     | f <sub>HCLK</sub><br>(MHz) | Typ   | Max                       |                           |                            |                            |                            | Unit |
|------------------------|------------------------------|--------------------------------|----------------------------|-------|---------------------------|---------------------------|----------------------------|----------------------------|----------------------------|------|
|                        |                              |                                |                            |       | T <sub>J</sub> =<br>30 °C | T <sub>J</sub> =<br>90 °C | T <sub>J</sub> =<br>110 °C | T <sub>J</sub> =<br>130 °C | T <sub>J</sub> =<br>140 °C |      |
| I <sub>DD(SLEEP)</sub> | Supply current in Sleep mode | All peripheral clocks disabled | 144                        | 2.05  | 2.35                      | 3.95                      | 5.50                       | 8.05                       | 10.00                      | mA   |
|                        |                              |                                | 48                         | 0.93  | 1.15                      | 2.80                      | 4.35                       | 6.95                       | 8.85                       |      |
|                        |                              | All peripheral clocks enabled  | 144                        | 15.00 | 15.50                     | 17.00                     | 18.50                      | 21.00                      | 22.50                      |      |
|                        |                              |                                | 48                         | 5.30  | 5.60                      | 7.00                      | 8.40                       | 11.00                      | 12.50                      |      |